From 2D TQFT to Gromov–Witten Theory

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Roadmap

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Why start with 2D TQFT?

- The Atiyah–Segal viewpoint packages cutting and gluing into a functorial formalism.
- In dimension 2, the theory becomes rigid enough to classify completely, yet rich enough to motivate later geometry.
- The central slogan is:

$$Z : \mathrm{Bord}^{SO}_{\langle 1,2 \rangle} \longrightarrow \mathrm{Vect}_k.$$

- The talk will use 2D TQFT as the clean algebraic model, and then explain why Gromov–Witten theory forces us to upgrade it.

2D TQFT AS A SYMMETRIC MONOIDAL FUNCTOR

Geometry

- Objects: closed oriented 1-manifolds
- Morphisms: oriented cobordisms
- Composition: gluing along common boundary
- Monoidal product: disjoint union

Algebra

- $Z(\emptyset) = k$
- $Z(M_0 \sqcup M_1) \cong Z(M_0) \otimes Z(M_1)$
- $Z(W_2 \circ W_1) = Z(W_2) \circ Z(W_1)$
- Orientation reversal corresponds to duality

Key idea

A topological operation on surfaces must become a linear-algebraic operation on state spaces.

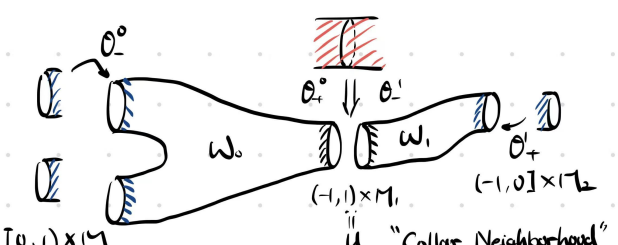

Figure. Gluing of cobordisms.

Elementary cobordisms generate the algebra

Let

$$A := Z(S^1).$$

Then the basic bordisms produce the structure maps

$$\eta : k \to A, \qquad \mu : A \otimes A \to A,$$

$$\Delta : A \to A \otimes A, \qquad \Theta : A \to k.$$

- Cup $\rightsquigarrow$ unit η
- Pair-of-pants $\rightsquigarrow$ multiplication μ
- Reversed pair-of-pants $\rightsquigarrow$ comultiplication Δ
- Cap $\rightsquigarrow$ trace Θ

The induced pairing is

$$\langle a, b \rangle := \Theta\big(\mu(a, b)\big).$$

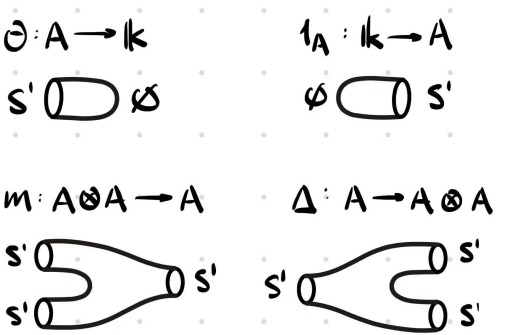

Figure. Fundamental components of 2D TQFT

Classification theorem

Theorem 1 (Abrams–Kock folklore theorem)

Oriented 2-dimensional TQFTs over k are equivalent to finite-dimensional commutative Frobenius algebras over k.

Why this is plausible:

- Any cobordism can be decomposed into cups, caps, and pairs-of-pants via Morse theory.
- Changing the decomposition gives Cerf moves.
- The corresponding algebraic identities are exactly associativity, commutativity, the unit law, and the Frobenius relation.

Moral

Once $A = Z(S^1)$ is known, the whole 2D TQFT is determined.

What 2D TQFT computes — and what it forgets

- Closed-surface amplitudes depend only on topological type, hence only on genus.
- This is ideal for pure cut-and-paste topology.
- But in enumerative geometry, correlation functions vary with:
  - complex structure of the source curve,
  - marked points,
  - degenerations to nodal curves,
  - cohomology insertions and descendant classes.
- Therefore the Atiyah–Segal axioms are too coarse for Gromov–Witten theory.

We need a field theory that lives over moduli spaces of curves, not just on diffeomorphism classes of surfaces.

WHY MODULI OF CURVES APPEAR

In Gromov–Witten theory, the domain is a marked Riemann surface

$$(C; p_1, \ldots, p_n).$$

- These domains vary in families, so the natural parameter space is $\mathcal{M}_{g,n}$.
- Noncompactness appears when necks pinch and smooth curves degenerate to nodal curves.
- Local model of smoothing a node:

$$zw = q, \qquad q \neq 0 \text{ smooth}, \qquad q \to 0 \text{ gives } zw = 0.$$

- Hence, compactification must allow singular but controlled limits.

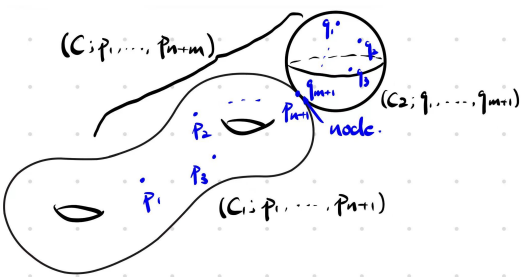

Figure. Guling two marked curves to generate a nodal curve.

Stable curves and $\overline{\mathcal{M}}_{g,n}$

- A prestable curve is a connected nodal curve with distinct marked smooth points.
- Stability requires every irreducible component to satisfy

$$2g_v - 2 + n_v > 0,$$

  where n_v counts marked points and branches of nodes on that component.
- The Deligne–Mumford compactification $\overline{\mathcal{M}}_{g,n}$ adds precisely these stable nodal limits.

Three fundamental morphisms

Forgetful maps, separating gluing, and non-separating gluing are the geometric operations that later become the CohFT axioms.

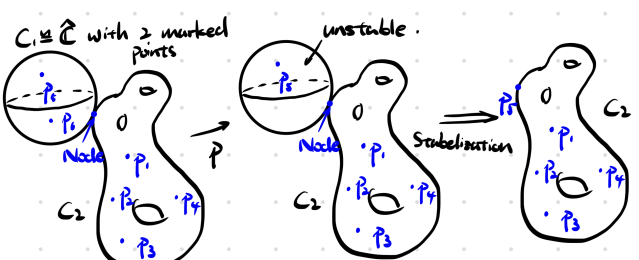

Figure. Forgetful map and stabilization

Stable graphs organize the boundary

A stable graph Γ records the combinatorics of a stable curve:

- vertices $\leftrightarrow$ irreducible components,
- genus label on each vertex,
- edges $\leftrightarrow$ nodes,
- legs $\leftrightarrow$ marked points.

The boundary stratification is indexed by graph type:

$$\overline{\mathcal{M}}_{g,n} = \bigsqcup_{\Gamma} S_\Gamma, \qquad \overline{S}_\Gamma = \mathrm{im}(i_\Gamma).$$

- For $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, the boundary consists of three special points.
- This is the geometric source of associativity in genus zero.

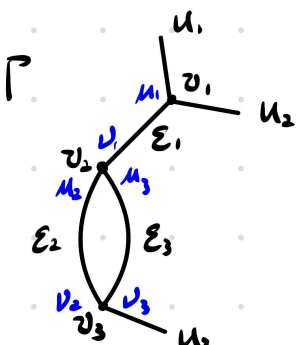

Figure. A stable graph representing a boundary case in $\overline{\mathcal{M}}_{1,3}$.

From TQFT to CohFT

2D TQFT

$$Z(W) : Z(M_0) \to Z(M_1)$$

outputs numbers or linear maps .

Depends only on topological bordisms.

CohFT

$$\Omega_{g,n} : V^{\otimes n} \to H^*(\overline{\mathcal{M}}_{g,n})$$

outputs cohomology classes on moduli space .

Remembers marked points, degenerations, and cohomological insertions.

Upgrade

CohFT is the moduli-aware refinement of 2D TQFT.

Axioms of a CohFT

The input is a state space $(V, \eta, \mathbf{1})$ with a nondegenerate symmetric bilinear form.
A CohFT is a system of multilinear maps

$$\Omega_{g,n} : V^{\otimes n} \longrightarrow H^*(\overline{\mathcal{M}}_{g,n}), \qquad 2g - 2 + n > 0,$$

satisfying:

- **Symmetry:** S_n-equivariance in the markings.
- **Gluing:** pullback to boundary strata equals contraction by η^{-1}.
- **Unit:** insertion of $\mathbf{1}$ matches the forgetful map, and

$$\Omega_{0,3}(v_1, v_2, \mathbf{1}) = \eta(v_1, v_2).$$

Schematically,

$$q^*\Omega_{g,n} = \sum_{i,j} \eta^{ij}\, \Omega_{g-1,n+2}(\cdots, e_i, e_j),$$

with an analogous formula for separating gluing.

Genus zero already gives a product

Since

$$\overline{\mathcal{M}}_{0,3} = \mathrm{pt},$$

the class $\Omega_{0,3}$ is just a trilinear form. Define the product by

$$\eta(a \bullet b, c) = \Omega_{0,3}(a, b, c).$$

- Symmetry gives commutativity.
- The unit axiom gives the identity element.
- Boundary factorization on $\overline{\mathcal{M}}_{0,4}$ gives the WDVV relation, hence associativity.

The degree-zero part

$$\omega_{g,n} := [\Omega_{g,n}]_0$$

is a genuine 2D TQFT, determined completely by $\Omega_{0,3}$.

Moduli of stable maps

Fix a smooth projective target X and a curve class $\beta \in H_2(X)$. The relevant moduli space is

$$\overline{\mathcal{M}}_{g,n}(X,\beta).$$

$$\begin{tikzcd} \overline{\mathcal{M}}_{g,n}(X,\beta) \arrow[r, "(\mathrm{ev}_1,\ldots,\mathrm{ev}_n)"] \arrow[d, "\pi"'] & X \times \cdots \times X \\ \overline{\mathcal{M}}_{g,n} & \end{tikzcd}$$

It comes with:

- evaluation maps $\mathrm{ev}_i : \overline{\mathcal{M}}_{g,n}(X,\beta) \to X$,
- a forgetful map $\pi : \overline{\mathcal{M}}_{g,n}(X,\beta) \to \overline{\mathcal{M}}_{g,n}$.

If everything were smooth of the expected dimension, one would define

$$\langle \gamma_1, \ldots, \gamma_n \rangle^X_{g,n,\beta} = \int_{\overline{\mathcal{M}}_{g,n}(X,\beta)} \prod_{i=1}^{n} \mathrm{ev}_i^*(\gamma_i).$$

The problem is that the moduli space is usually not this well-behaved.

Why the virtual fundamental class is necessary

The stable-map moduli space can be singular, have excess dimension, or fail to be of pure dimension.

The deformation theory is controlled by the linearization of the J-holomorphic equation

$$\bar{\partial}_J f = 0, \qquad D_f \bar{\partial}_J : \Gamma(C, f^* TX) \to \Omega^{0,1}(C, f^* TX).$$

The correct replacement for the actual dimension is the virtual dimension :

$$\mathrm{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{g,n}(X, \beta) = \int_\beta c_1(TX) + (1-g)(\dim_{\mathbb{C}} X - 3) + n.$$

Virtual class

One integrates against

$$[\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\mathrm{vir}},$$

a cycle of exactly the expected dimension.

Gromov–Witten theory as a CohFT

This is the conceptual completion of the manuscript:

$$\Omega^X_{g,n}(\gamma_1, \ldots, \gamma_n) := \sum_\beta q^\beta \, \pi_* \left([\overline{\mathcal{M}}_{g,n}(X, \beta)]^{\mathrm{vir}} \prod_{i=1}^n \mathrm{ev}_i^*(\gamma_i) \right).$$

- The target state space is $V = H^*(X)$ with the Poincaré pairing.
- Pushforward along π produces classes on $\overline{\mathcal{M}}_{g,n}$.
- Virtual gluing theorems imply the CohFT factorization axioms.
- Thus, Gromov-Witten theory is not merely analogous to TQFT, it is a canonical CohFT .

Quantum cohomology and semisimplicity

The genus-zero three-point invariants deform the ordinary cup product into the quantum product:

$$\eta(a * b, c) = \sum_{\beta} q^{\beta} \langle a, b, c \rangle^{X}_{0,3,\beta}.$$

- This produces the quantum cohomology ring $QH^*(X)$.
- The WDVV equation is the associativity constraint for $*$.
- Semisimplicity of the resulting Frobenius algebra is the gateway to structural results.
- In the semisimple case, higher genus is governed to a remarkable extent by genus-zero data.

This is the bridge from the Frobenius algebra of 2D TQFT to the Frobenius geometry of Gromov–Witten theory.

Frobenius manifolds: the genus-zero package

A Frobenius manifold is a manifold M equipped with

- a commutative, associative product $\circ$ on the tangent bundle TM,
- a flat metric η with a unit vector field e,
- and locally a potential Φ such that

$$c_{ijk} := \eta(\partial_i \circ \partial_j, \partial_k) = \frac{\partial^3 \Phi}{\partial t^i \, \partial t^j \, \partial t^k}.$$

Relation to Gromov–Witten theory

Let $\{T_i\}$ be a basis of $H^*(X)$, and write $t = \sum_i t^i T_i$. The genus-zero Gromov–Witten potential is

$$\Phi_0^X(t) = \sum_{n,\beta} \frac{q^\beta}{n!} \langle t, \ldots, t \rangle_{0,n,\beta}^X .$$

Its third derivatives define the product on each tangent space:

$$\partial_i \circ_t \partial_j = \sum_{k,\ell} \eta^{k\ell} \frac{\partial^3 \Phi_0^X}{\partial t^i \, \partial t^j \, \partial t^\ell} \, \partial_k.$$

The WDVV equation is exactly the associativity condition for $\circ_t$.

Conclusion

Main message

The passage from 2D TQFT to Gromov–Witten theory is a passage from topological cutting-and-gluing to moduli-level factorization : The output is no longer just a scalar or a linear map, but a cohomology class on $\overline{\mathcal{M}}_{g,n}$.

- Oriented 2D TQFTs are classified by finite-dimensional commutative Frobenius algebras.
- $\overline{\mathcal{M}}_{g,n}$ provides the natural geometric setting for degeneration, stabilization, and gluing.
- CohFT refines TQFT by replacing numerical amplitudes with classes on moduli spaces.
- Gromov–Witten theory is the central example of this refinement.
- In genus zero, this produces quantum cohomology; semisimplicity leads toward the Givental–Teleman picture.

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